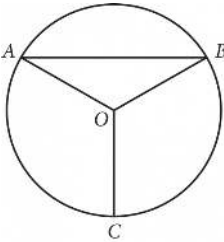


Question ID: 69b0d79d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question



Point O is the center of the circle above, and the measure of $\angle OAB$ is 30° . If the length of $\overline{OC}$ is 18, what is the length of arc $\widehat{AB}$?

Answer

- A. 9π
- B. 12π
- C. 15π
- D. 18π

Correct Answer: B

Rationale

Choice B is correct. Because segments OA and OB are radii of the circle centered at point O, these segments have equal lengths. Therefore, triangle AOB is an isosceles triangle, where angles OAB and OBA are congruent base angles of the triangle. It's given that angle OAB measures 30° . Therefore, angle OBA also measures 30° . Let x° represent the measure of angle AOB. Since the sum of the measures of the three angles of any triangle is 180° , it follows that $30^\circ + 30^\circ + x^\circ = 180^\circ$, or $60^\circ + x^\circ = 180^\circ$. Subtracting 60° from both sides of this equation yields $x^\circ = 120^\circ$, or $\frac{2\pi}{3}$ radians. Therefore, the measure of angle AOB, and thus the measure of arc $\widehat{AB}$, is $\frac{2\pi}{3}$ radians. Since $\overline{OC}$ is a radius of the given circle and its length is 18, the length of the radius of the circle is 18. Therefore, the length of arc $\widehat{AB}$ can be calculated as $\left(\frac{2\pi}{3}\right)(18)$, or 12π .

Choices A, C, and D are incorrect and may result from conceptual or computational errors.